BTS–IV(R/S)–06–22–0794 Reg. No. 201120032

C

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

CE 19-200-0401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Transform a region to another using conformal mapping.
CO2: Evaluate real integrals using Residue theorem.
CO3: Form and solve partial differential Equations.
CO4: Determine solution of partial differential equation for vibrating string and heat conduction.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A

(Answer *ALL* questions)

		Question	Marks	BL	CO	PO
I.		$(8 \times 3 = 24)$				
	(a)	Show that an analytic function with constant modulus is constant.	3	2	1	2
	(b)	Show that by the transformation $W = Z + \frac{1}{Z}$ the circles in the Z-plane $\lvert z\rvert = r\ (r \neq 1)$ transformed into a family of ellipse of W – plane.	3	2	1	2
	(c)	Evaluate f(2) and f(3) if $f(a) = \int_C \frac{2z^2 - z - 2}{z - a} dz$ where C is the circle $\lvert z\rvert = \frac{5}{2}$.	3	3	2	3
	(d)	Evaluate using Cauchy's integral formula $\int_C \frac{3z+5}{z^2+2z} dz$ where C is the circle $\lvert z\rvert = 1$.	3	3	2	3
	(e)	Solve: $p + q = Z$.	3	2	3	2
	(f)	Solve: $p^2 y(1 + x^2) = qx^2$.	3	2	3	2
	(g)	Derive one dimensional heat equation.	3	1	4	1
	(h)	Solve the equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$ given $u(x, 0) = 6e^{-3x}$.	3	2	4	2

PART B

$(4 \times 12 = 48)$

		Question	Marks	BL	CO	PO
II.	(a)	If f(z) is a regular function of z prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\lvert f(z)\rvert^2 = 4\lvert f'(z)\rvert^2$	6	3	1	3
	(b)	Show the transformation $W = z + \frac{1}{z}$ converts the straight line $\arg z = \theta$ when $\lvert\theta\rvert < \frac{\pi}{2}$ in to a branch of the hyperbola of eccentricity $\sec\theta$.	6	2	1	2

OR

(P.T.O.)

BTS–IV(R/S)–06–22–0794

III.	(a)	If f(z) = u + iv is an analytic function find f(z) if $u + v = \frac{x}{x^2+y^2}$ when f(1) = 1.	6	3	1	3
	(b)	Find the bilinear transformation which maps the points z = 0, i, -1 on to the point w = i, 0, ∞ respectively in w – plane. Also find the fixed points.	6	2	1	2
IV.	(a)	State Cauchy's integral formula and using it evaluate $\int_c \frac{z^2 dz}{(z-1)^2(z+2)}$ where C is the circle $\lvert z \rvert = 3$.	6	1, 3	2	3
	(b)	Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent's series valid in the region $0 < \lvert z+1 \rvert < 2$ and find the residue of f(z) at z = -1.	6	2	2	2
		OR				
V.	(a)	State Cauchy's Residue theorem and using it evaluate $\int_c \frac{e^z}{z^2+\pi^2} dz$ when c is the circle $\lvert z \rvert = 4$.	6	1, 3	2	3
	(b)	Evaluate $\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta}$ around the unit circle $\lvert z \rvert = 1$.	6	2	2	2
VI.	(a)	Form a partial differential equation by eliminating the arbitrary function f and g from z = yf(x) + xg(y)	6	2	3	2
	(b)	Solve: r + s – 6t = y cos x.	6	3	3	3
		OR				
VII.	(a)	From a partial differential equation by eliminating arbitrary function F from $F(xy + z^2, x + y + z) = 0$.	6	2	3	2
	(b)	Solve: $x(y^2 - z^2)p - y(z^2 + x^2)q = z(x^2 + y^2)$.	6	3	3	3
VIII.	(a)	Solve by the method of separation of variables $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$.	6	4	4	4
	(b)	Derive D'Alembert's solution of the wave equation.	6	1	4	1
		OR				
IX.	(a)	Derive the solution of one dimensional heat equation by the method of separation of variable.	6	1	4	1
	(b)	An insulated rod of length *l* has its ends A and B maintained at $0^{\circ}C$ and $100^{\circ}C$ respectively until study state conditions prevail. If B is suddenly reduced to $0^{\circ}C$ and maintained at $0^{\circ}C$, find the temperature at a distance x from A at time t.	6	4	4	4

Bloom's Taxonomy Levels
L1 = 15%, L2 =45%, L3 = 25%, L4 = 15%

***